

Proposal for a Ghana RPAS Practical Training, Qualification and Digital Assessment Framework

A competency-based, system-assessed model for RPAS pilot training, practical testing and training-organisation oversight

Field	Description
Prepared for	Government of Ghana, Ghana Civil Aviation Authority (GCAA), and relevant partner institutions
Proposal status	Pre-approval technical proposal for discussion. Not an official GCAA directive or approved qualification standard.
Core basis	GCAA RPAS framework, GCAD Part 28, Ghana-specific pilot training concept, and v3 radius-based RTK practical assessment model
Budget basis	15-trainee sequential course model: classroom → simulator → practical flight → RTK automated assessment
Recommended planning budget	USD \$127,380 (equipment + RTK automated assessment system + contingency + import/logistics allowance)
Date	24 July 2026

Important note: the qualification names, practical-test dimensions, scoring model and budget are proposed design elements. Official names, legal effect, licensing authority and implementation conditions are to be confirmed by GCAA.

1. Executive Summary

This proposal sets out a Ghana-focused RPAS practical training, qualification and digital assessment framework. It is designed to support a transition from time-based and manual training records toward competency-based assessment using online theory testing, simulator logs, RTK flight-data capture and structured training-organisation oversight.

The proposed Basic Pilot practical standard is built around an 8-figure practical flight task. The planned flight path is assessed through a radius-based error model: the scoring zones are measured from the nominal path centreline, not by corridor diameter.

The proposed system also introduces a scalable qualification pathway: RPAS Basic Pilot, RPAS Professional Pilot endorsements, RPAS Instructor, RPAS Training Manager / Safety Supervisor, Training Organisation Approval and Operator Certificate integration. These names are proposal labels and must be aligned with GCAA's official RPL, ROC, RTOC and related authorisation structures before implementation.

For initial deployment, the recommended operating model is a 15-trainee sequential course: classroom instruction, simulator training, practical flight training and RTK automated assessment. The recommended planning budget is USD \$127,380, excluding VAT, customs duties, Ghana inland transport, insurance, GCAA fees, facility fit-out, instructor fees, airfare and lodging.

Design pillar	Proposed approach	Expected result
Competency-based	Use simulator tasks, online tests and RTK flight data rather than relying primarily on logged flight hours.	More objective evidence of competence.
System-assessed	Use automated scoring and data integrity controls to reduce subjective examiner discretion.	Greater consistency and auditability.
GCAA-aligned	Use GCAA RPAS categories and existing Part 28 terminology as the regulatory reference point.	Easier policy discussion and phased adoption.
Institutional quality	Tie training approval to instructors, safety supervisors, records, manuals and safety controls.	Improved training-organisation oversight.

1.1 Status of information

Type	Meaning in this proposal	Examples
Official basis	Items reflected from GCAA publications or current GCAA RPAS resources.	RPAS = RPA + RPS + C2; all unmanned aircraft are subject to GCAD Part 28; GCAA RPAS Safety Training is linked to registration.
Proposal	Design choices recommended for the Ghana pilot model but not yet confirmed as GCAA official test standards.	Rcourse 5m/7m/10m, rerror 1m/2m/3m scoring zones, 70-point pass threshold.

For GCAA confirmation	Items that require formal review, approval or calibration with GCAA and local stakeholders.	Official qualification names, certificate authority, GPS/ATTI policy, weather minima, retest rules, data retention period.
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2. Regulatory and institutional basis

This proposal is designed to sit within Ghana's existing RPAS framework rather than replace it. The following points are treated as official reference points for discussion.

Official basis	Implication for this proposal	Source
GCAA RPAS concept	GCAA describes RPAS as a system consisting of the Remotely Piloted Aircraft (RPA), Remote Pilot Station (RPS), and Command and Control (C2) link.	[1]
Applicability of Part 28	GCAA guidance states that unmanned aircraft, whether piloted, autonomous or a combination, are subject to GCAD Part 28.	[1]
RPAS classification	Part 28 classifies RPAS as Small (MTOW up to 1.5 kg), Light (more than 1.5 kg and up to 7 kg), and Large (more than 7 kg).	[2]
Training and commercial operation controls	Part 28 includes RTOC and ROC concepts, RPL-related requirements, 400 ft AGL and VLOS/BVLOS-related controls.	[2]
Existing safety training	GCAA RPAS Safety Training is presented as a prerequisite for drone registration and is structured into three modules with an 80% pass mark and a certificate valid for one year.	[3]
RPAS application ecosystem	GCAA RPAS Resources include applications for RPAS import, use, BVLOS/above 400 ft AGL operation, RPAS Pilot Licence and other related processes.	[4]

PROPOSAL The proposed training and assessment model should be submitted as a pilot framework for GCAA review, not as a replacement for existing permits, registrations, RPL, ROC or RTOC requirements.

FOR GCAA CONFIRMATION The relationship between the proposed Basic Pilot, Professional Pilot and Instructor designations and GCAA's official licensing/authorisation framework should be formally mapped before launch.

3. Proposed Ghana RPAS qualification pathway

Stage	Proposed designation	Purpose	Core requirements
1	RPAS Basic Pilot	Low-risk VLOS pilot qualification for entry-level and recreational users.	Online theory assessment, simulator tasks, 8-figure RTK practical assessment.
2	RPAS Professional Pilot	Operational endorsement for sector-specific work.	Basic Pilot, sector training, scenario-based task evaluation.

3	RPAS Instructor	Training qualification for Basic and Professional courses.	Professional Pilot endorsement, one-year operational experience, instructor training and teaching assessment.
4	Training Manager / Safety Supervisor	Training quality and safety oversight role.	Professional background, instructor experience, safety management course and assessment.
5	Training Organisation Approval	Institutional approval for professional RPAS education.	Instructor, safety supervisor, facilities, equipment, manuals, records and safety controls.
6	Operator Certificate integration	Commercial and public operator pathway.	Organisation, equipment, SOPs, SMS, insurance and operational capability review.

FOR GCAA CONFIRMATION These names are proposed operational labels. Their legal effect, certificate wording and equivalence to existing GCAA instruments must be decided by GCAA.

3.1 Professional Pilot endorsement examples

Sector	Example endorsement	Operational focus
Agriculture	Professional Pilot - Agriculture	Spraying support, crop monitoring, pest surveillance and plant-health assessment.
Mapping	Professional Pilot - Mapping	Aerial photography, orthomosaic capture, 3D mapping and ground-control workflows.
Inspection	Professional Pilot - Inspection	Powerline, bridge, solar farm and industrial asset inspection.
Public Safety	Professional Pilot - Public Safety	Disaster response, search support, rescue support and incident reporting.
Maritime / Environment	Professional Pilot - Maritime / Environmental Monitoring	Coastal, water-quality, forest and environmental monitoring.
Security Surveillance	Professional Pilot - Security Surveillance	Perimeter monitoring, illegal-mining surveillance and critical-facility monitoring.
Drone Sports / Education Support	Professional Pilot - Drone Sports / Education Support	Drone sports facilitation, public outreach and education assistance.

4. Practical flight-test and RTK assessment standard

The Basic Pilot practical standard is centred on a figure-8 flight course and a radius-based automated scoring method. The course radius (Rcourse) and scoring error radius (rerror) are separate variables.

Variable	Meaning	Proposed values	Note
P	Pilot position	15 m behind take-off point T	Common baseline for all classes.
T	Take-off and landing point	Start/return reference point	Can be linked to the RTK reference coordinate system.
C	Actual figure-8 crossing point	T to C = Rcourse	The figure-8 path crosses at C.
Rcourse	Course circle radius	Small 5 m / Light 7 m / Large 10 m	Course geometry parameter.
rerror	Scoring error radius	1 m / 2 m / 3 m	Measured from the planned path centreline. Not diameter-based.

GCAA class	MTOW basis	Operating basis	Proposed Rcourse	Design purpose
Small RPAS	≤ 1.5 kg	VLOS	5 m	Basic control, position and altitude stability.
Light RPAS	> 1.5 kg to ≤ 7 kg	VLOS	7 m	Inertia, speed and turning-radius control.
Large RPAS	> 7 kg	VLOS or BVLOS with prior authorisation	10 m	Safety distance, braking, emergency procedures.

4.1 Radius-based auto scoring

Zone	error criterion	Meaning	Sample score	Assessment interpretation
A Zone	$r \leq 1$ m	Target corridor	100	Precise flight.
B Zone	$1 \text{ m} < r \leq 2$ m	Minor deviation	80	Passable control.
C Zone	$2 \text{ m} < r \leq 3$ m	Major deviation	60	Borderline performance.
D Zone	$r > 3$ m	Course deviation	0	Safety cap and auto-fail rules apply.

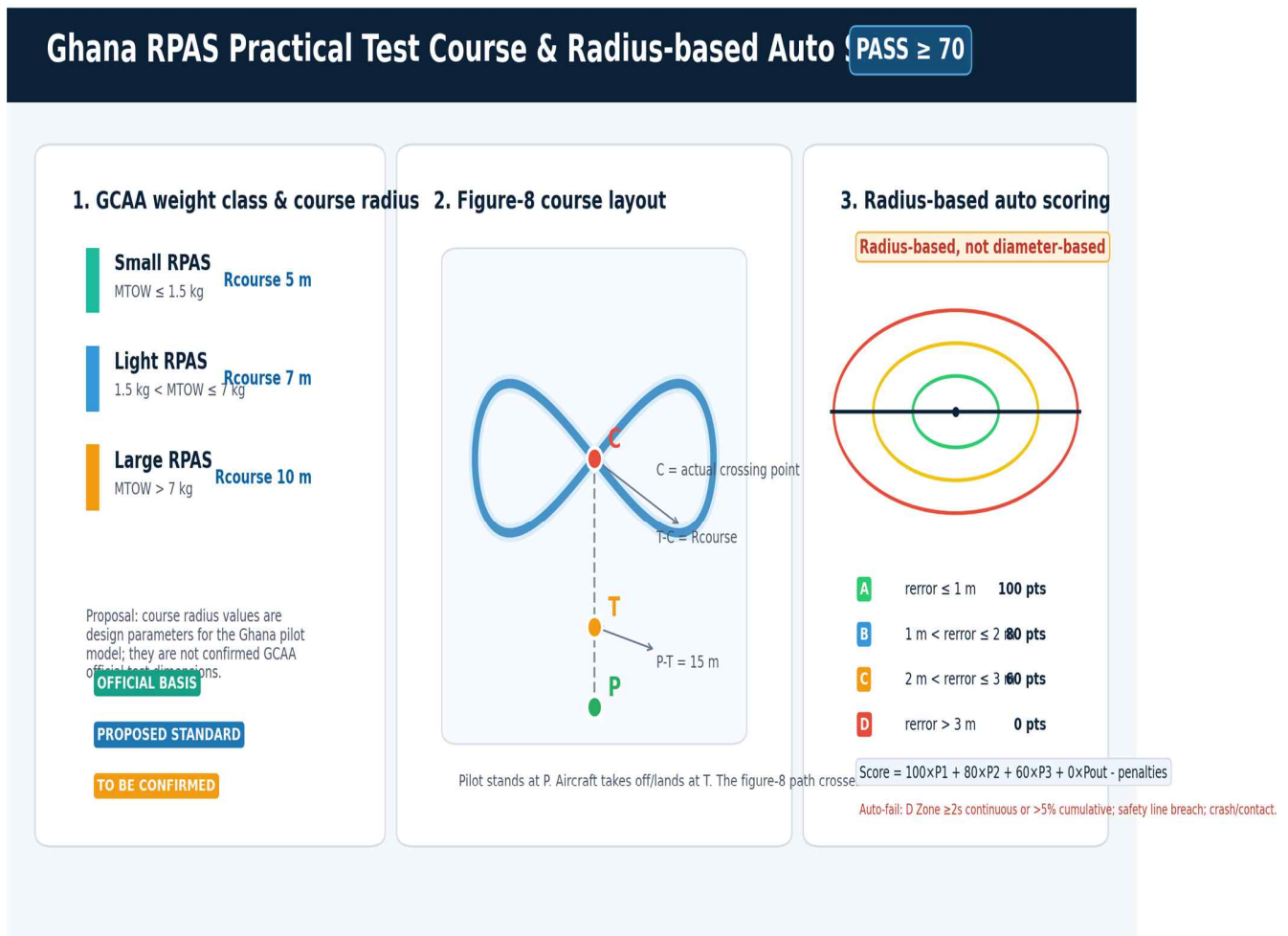
Proposed formula: Final Score = $100 \times P_1 + 80 \times P_2 + 60 \times P_3 + 0 \times P_{out}$ - safety/procedure penalties

P_1 , P_2 , P_3 and P_{out} represent the proportion of samples within the A, B, C and D zones respectively. The passing score is proposed at 70 points or higher, provided that no automatic failure condition is triggered.

Automatic failure / cap condition	Treatment	Purpose
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D Zone continuous for 2 seconds or more	Automatic failure	Prevents sustained course deviation.
D Zone cumulative time over 5% of the course	Automatic failure	Prevents unsafe flight from being hidden by average scoring.
Safety line or test-field boundary breach	Automatic failure	Protects people and facilities.
Missing crossing point, wrong direction or skipped sequence	Zero for section or automatic failure	Ensures course completion and sequence compliance.
Crash, rollover, obstacle contact or loss of control	Automatic failure	Maintains basic flight safety.
C Zone residence over 25%	Maximum score cap of 75	Controls repeated unstable deviations.
Any D Zone entry	Maximum score cap of 90	Reflects the seriousness of a course deviation.

Figure 1. Proposed practical test course and radius-based auto scoring standard.



Draft proposal | Course and scoring parameters are subject to GCAA review and pilot calibration.

5. Digital assessment system and data integrity

The proposed model minimises subjective examiner judgement by combining online theory testing, simulator logs and RTK practical flight data. The examiner's role is primarily to verify identity, check equipment, control field safety, start and end the test, and record exceptions.

The RTK practical assessment should record position, altitude, speed, path deviation, hover stability, waypoint/crossing-point achievement and landing error as source logs. The assessment result should be stored with the original data to support audit and recalibration.

Control item	Proposed requirement
Pre-test equipment check	RTK base/rover status, reference coordinate, Fix status, battery and storage device must be checked before each test.
Antenna offset	Each aircraft's RTK antenna mounting offset should be recorded and applied to scoring calculations.
Original logs	Raw logs should be preserved in a non-editable or audit-controlled format with the result report.
Signal loss / time sync	RTK signal loss, missing samples, time synchronisation error or reference-point misconfiguration should trigger retest or invalidation rules.
Pilot calibration	Weather limits, wind thresholds, sampling rate, error calculation method and retest rules should be calibrated through the first pilot cohort.

PROPOSAL The RTK automated assessment system is presented as one integrated line item in the budget. Product-level details and vendor confirmations should be handled in procurement documents, not in the policy proposal.

6. Training organisation approval and oversight

The proposed approval model treats an RPAS training organisation as an accountable education and safety system, not simply as a classroom with aircraft.

Area	Minimum proposed requirement
Personnel	At least one Instructor and one Training Manager / Safety Supervisor.
Facilities	Classroom, simulator area, briefing/debriefing area and maintenance-practice space.
Equipment	Training drones, simulator equipment, RTK assessment equipment and field safety equipment.
Documents	Training Manual, Operations Manual and Safety Manual.
Records	Trainee records, simulator logs, RTK assessment logs, maintenance records and incident records.

Safety	Emergency response plan, risk assessment template, accident/incident reporting procedure.
Oversight	Acceptance of GCAA inspection, renewal review and corrective-action follow-up.

6.1 Training Manager / Safety Supervisor

Subject	Weight	Purpose
RPAS safety management	25%	Hazard identification, risk matrix and emergency response.
Training-organisation management	20%	Training records, instructor management, equipment management and quality control.
Practical assessment safety control	20%	Field control, RTK assessment process and emergency stop criteria.
Accident and incident reporting	15%	Investigation, reporting structure and recurrence prevention.
Regulations and approval criteria	10%	GCAA approval, training organisation approval and renewal review.
Scenario-based judgement	10%	Decision-making in realistic training-organisation situations.

7. Initial 2026 Master Instructor Foundation Course

Item	Proposed content
Course name	Ghana RPAS Master Instructor Foundation Course
Timing	Early September 2026, two-week course
Location	GATA 6th-floor training facility, subject to host confirmation
Participants	Approximately 15 candidates
Target audience	Manned-aircraft pilots, existing drone-qualified personnel, GCAA/GATA/ATU stakeholders and relevant public-sector participants.
Course purpose	Develop the initial instructor cohort while validating the Ghana RPAS training, assessment and qualification framework.
Module	Content
Regulation and institutions	Ghana RPAS framework, airspace, authorisations and training organisation approval.
System principles	Flight control, GPS/RTK, IMU, GCS, failsafe and battery systems.
Simulator	Basic flight, figure-8 flight, emergency procedures and mission flight.

Practical assessment	RTK-based figure-8 course and mission-type assessment.
Instructional method	Lesson design, briefing, flight coaching and debriefing.
Assessment operation	Online theory test, simulator logs and RTK assessment-data management.
Safety management	Risk assessment, incident reporting, emergency response and safety supervision.
Framework design	Qualification framework, training organisation criteria and assessment-procedure drafts.

7.1 Expected pilot-course outputs

- Ghana RPAS Digital Assessment Model v0.1
- Ghana RPAS Qualification Framework v0.1
- 8-Figure RTK Practical Test Standard v0.1
- Professional Pilot Endorsement Framework v0.1
- Instructor Training Manual v0.1
- Training Manager / Safety Supervisor Standard v0.1
- Training Organisation Approval Criteria v0.1
- Online Theory Test Bank v0.1
- Simulator Training Module v0.1

8. Implementation roadmap

Phase	Core activity	Main outputs
1	Train the initial instructor cohort using GATA facilities.	Instructor candidates, framework draft and assessment draft.
2	Pilot the Basic Pilot course with online theory, simulator tasks and figure-8 RTK practical assessment.	Calibrated difficulty, tolerances and pass criteria.
3	Open sector-specific Professional Pilot endorsement courses.	Agriculture, mapping, inspection, public safety and other endorsement standards.
4	Institutionalise Instructor and Training Manager / Safety Supervisor pathways.	Mandatory staffing criteria for training organisations.
5	Approve and scale professional training organisations.	Accra, Kumasi and hub-and-spoke expansion model.
6	Prepare regional extension across West Africa.	ECOWAS and African technical-university network model.

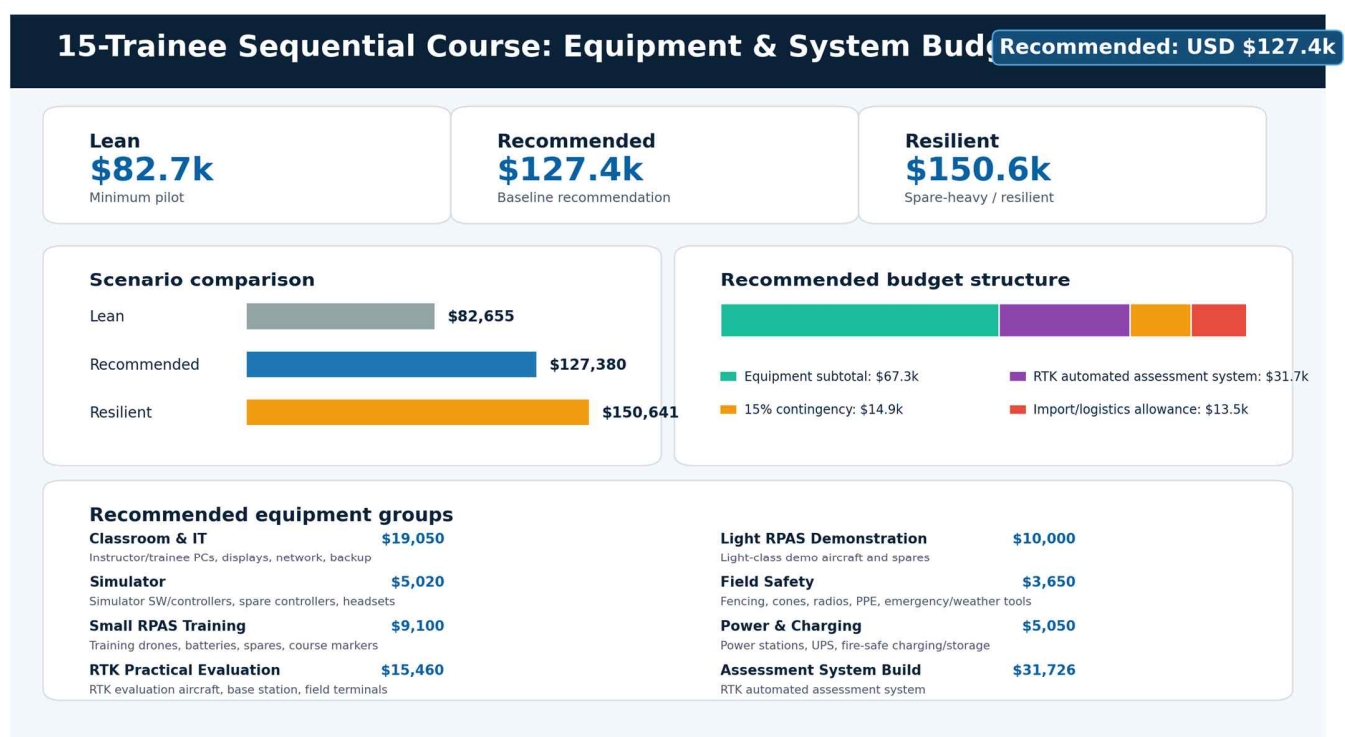
9. Equipment and budget plan for the 15-trainee sequential model

The budget model assumes one 15-trainee cohort moving sequentially through classroom instruction, simulator training, practical flight training and RTK automated assessment. The RTK automated assessment system is kept as one consolidated budget line.

Scenario	Equipment subtotal	RTK automated assessment system	Subtotal	15% contingency	Import/logistics allowance	Total planning budget
Lean	\$34,200	\$31,726	\$65,926	\$9,889	\$6,840	\$82,655
Recommended	\$67,330	\$31,726	\$99,056	\$14,858	\$13,466	\$127,380
Resilient	\$84,560	\$31,726	\$116,286	\$17,443	\$16,912	\$150,641

RECOMMENDATION Adopt the Recommended scenario as the baseline pilot budget: USD \$127,380. The Lean scenario is suitable only for minimum proof-of-concept operation, while the Resilient scenario provides stronger spare capacity and failure tolerance.

Figure 2. Recommended budget structure and equipment groups.



Planning budget excludes VAT, customs duties, Ghana inland transport, insurance, GCAA fees, facility fit-out, instructor fees, airfare and lodging.

9.1 Recommended equipment groups

Equipment group	Main components	Recommended amount
Classroom & IT	Instructor and trainee PCs, displays, network and backup storage.	\$19,050
Simulator	Simulator software/controller sets, spare controllers and headsets.	\$5,020
Small RPAS Training	Small training aircraft, batteries, consumables and course markers.	\$9,100
RTK Practical Evaluation	RTK evaluation aircraft, RTK base station, field tablets and log accessories.	\$15,460
Light RPAS Demonstration	Light-class demonstration aircraft, batteries and maintenance spares.	\$10,000
Field Safety	Safety fencing, cones, radios, PPE, emergency and weather equipment.	\$3,650
Power & Charging	Portable power, UPS, fire-safe charging/storage and cabling.	\$5,050
Assessment System Build	Consolidated RTK automated assessment system line item.	\$31,726

9.2 Budget exclusions and procurement controls

The planning budget excludes VAT, customs duties, Ghana inland transport, insurance, GCAA fees, facility fit-out, instructor fees, airfare and lodging. These should be added during procurement and implementation planning.

Control	Proposed procurement treatment
Vendor quotation	Obtain updated local or regional quotations before purchase, especially for RTK equipment and software integration.
System acceptance	Conduct field calibration and acceptance testing with sample flight data before using automated results for official decisions.
Data integrity	Require raw log retention, audit trail and standardised result reports.
Scalability	Separate one-time development/configuration costs from per-set hardware/software quantities.
Regulatory clearance	Confirm import, registration, use and pilot licensing requirements through GCAA before equipment shipment or field use.

10. Items for GCAA confirmation

Topic	Decision to be confirmed	Why it matters
Official approval scope	Qualification names, training organisation criteria and certificate/approval issuing authority.	Prevents confusion with existing RPL, ROC, RTOC and permit structures.
Existing RPAS Safety Training	Whether it is a prerequisite, common module or recognised equivalent for the Basic Pilot pathway.	Avoids duplication and aligns with current registration training.
RTK assessment criteria	Tolerance values, altitude criteria, sampling rate, standard aircraft and retest thresholds.	Ensures defensible and repeatable scoring.
GPS/ATTI policy	Whether GNSS-aided flight, ATTI/manual mode or class-based differentiation should apply.	Aligns skills assessment with safety and equipment realities.
Weather minima	Wind, visibility, rain, night-operation and test-interruption thresholds.	Prevents unsafe or incomparable test conditions.
Online test security	Initial online-form controls and long-term GCAA LMS/portal requirements.	Protects examination integrity.
Training organisation staffing	Initial dual-hatting allowance and timeline for separating instructor and safety supervisor roles.	Supports phased expansion without compromising oversight.
High-risk operations	BVLOS, night flight, delivery, security surveillance, counter-drone and military/police missions.	Keeps high-risk activities outside the initial general qualification until separate controls are approved.

11. Expected benefits

- Improved fairness and transparency in RPAS practical assessment by reducing dependence on manual flight-hour logs and examiner discretion.
- A stronger training-organisation quality-management model using instructors, safety supervisors, manuals, records and GCAA oversight.
- A national RPAS training infrastructure that can use GATA training space for education, simulation, assessment and certification preparation.
- Sector-specific workforce development for agriculture, public safety, maritime monitoring, environmental monitoring, mapping and infrastructure inspection.
- A platform for regional extension through ECOWAS and African technical-university networks after successful pilot validation in Ghana.

12. References and source basis

Re f.	Source	URL / note
[1]	GCAA, Remotely Piloted Aircraft Systems / Drones guidance page	https://www.gcaa.com.gh/web/remotely-piloted-aircraft-system-2/
[2]	Ghana Civil Aviation (Remotely Piloted Aircraft Systems) Directives, 2018, Part 28	https://www.gcaa.com.gh/web/wp-content/uploads/2018/directives/GCA_FLIGHT_STANDARDS_DIRECTIVE/PART_28_REMOTELY_PILOTED_AIRCRAFT_SYSTEMS_DIRECTIVES.pdf
[3]	GCAA RPAS Safety Training Portal	https://www.gcaa.com.gh/rpas/
[4]	GCAA RPAS Resources page	https://www.gcaa.com.gh/web/rpas-resources/
[5]	User-provided source draft	Ghana RPAS Digital Assessment & Qualification Framework v3 / Korean v3 document uploaded for this proposal
[6]	User-provided budget workbook	Ghana_RPAS_15pax_Sequential_Budget_Simplified_v3.1_260724.xlsx

Ghana RPAS Digital Assessment & Qualification Framework | Submission Proposal
End of proposal